THE UNIVERSITY OF WINNIPEG

FINAL PROJECT
AI STATEMENT

Course: Current Topics in Computing (CAS)

Instructor: Professor Sergio G. Camorlinga

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TOOLS USED

- Claude.

EXACT PROMPTS

- Act like a strict tutor from graduate program in computer science major. Help me brainstorm, draft outlines, and check my research report about the topic: "Frailty Syndrome in Older Adults: Modeling as a Complex Adaptive System." The project will be a network model written in Python, focusing mostly on the self-organization and emergence of the system.
- Suggest the optimal order to read through the papers.
- Based on the paper by Fried "Frailty in Older Adults: Evidence for a Phenotype", help me synthesize and generate the dataset containing the following requirements:
    - 10% error according to table 1
    - 200 patients with index and gender, bmi and height
    - 4 main fields: weakness, slowness, low_activity, exhaustion
    - according to the appendix, classify each patient to the correct state
- After trying the original parameters from Mitnitski&Rockwood, my results cannot match expectations, help me test out and suggest several relevant parameters:
    - C0, repair resistance, repair and damage rate.
- Those are my plots, help me align them, and optimize for better visual, and give me the global font, can be reusable to implement on graphs. Check if there are any points that needs improvement.
- Based on the given template and the IEEE format, check if there are any parts that need to be adjusted.
- I have done my report, check punctuation, grammar and vocabulary. Suggest which part can be polished but keeping the same original tone.

OUTPUT CHECK

- Read the papers multiple times and highlight key points that could be relevant to the project requirements.
- After obtaining the parameters, I did quite a lot of research across different papers to verify if the values are acceptable, and if they follow some source of rules. Since the public datasets are very limited, this is also a limitation of this project's dataset.
- Developed the base code flow and the algorithm.
- Performed error checking of the source code.
- Verify model outputs to check if they correlated to my own hypothesis.

REFLECTION ON LEARNING

- AI can make mistakes, and outputs can be incorrect. Outputs should be always verified.
- AI is a useful tool to improve learning performance, not to rely on entirely.